

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' —symptom: blank screen, no POST beep).

Hardware Issue	Typical Symptom
1. RAM failure	PC may not boot, show no display, or produce repeated beep codes.
2. Power supply failure	PC does not turn on, or shuts down unexpectedly.
3. Hard drive/SSD failure	Slow boot, operating system errors, or files becoming inaccessible.
4. CPU overheating	PC becomes very hot, runs slowly, or shuts down/restarts automatically.
5. Graphics card (GPU) failure	No display, screen artifacts, flickering, or crashes during graphics-intensive tasks.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Laptop Not Charging — Step-by-Step Troubleshooting

Step	What to Check	What to Look For
1. Check the power outlet	Plug another device into the same outlet.	Make sure the outlet is supplying power.
2. Inspect the charger	Check the adapter, cable, and plug for damage or loose connections.	Cuts, bent pins, burning marks, or loose wires.
3. Check the charging port	Look inside the laptop's charging port.	Dust, bent pins, or a loose port.
4. Try another compatible charger	Use a known-working charger with the correct voltage/current rating.	If it charges, the original charger is likely faulty.
5. Test the charger with a multimeter	Measure the DC output voltage of the charger.	Compare the reading with the voltage printed on the adapter.
6. Check the battery connection	If accessible, inspect the battery connector.	Make sure the connector is firmly seated and there is no visible damage.

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

For a **standard 5V USB charger**, a sample multimeter test can be recorded like this:

Test	Result
Charger rated output	5.0 V DC
Multimeter measured voltage	5.08 V DC
Expected range	4.75–5.25 V
Result	Within expected range — Charger output is normal

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

12V Power Supply Test

1. **Switch off and unplug the PC** from the wall outlet.
2. Disconnect the **ATX power connector** from the motherboard.
3. Set the multimeter to **DC voltage (V=)**.
4. Set the range to **20V DC** if the meter is not auto-ranging.
5. Put the **black probe on a black (GND) wire**.
6. Put the **red probe on the yellow (+12V) wire**.
7. Read and record the voltage. For example: **11.9V DC**.
8. Compare the reading with the expected **11.5–12.6V range** for this test.

Example result: 11.9V DC — Normal.

If the voltage is below 11.5V: it could indicate a **weak or failing power supply (PSU)** that is not providing stable 12V power, which can cause random shutdowns.